System Cu–Fe

Solution Phases:

(stable)	:	Liquid, fcc–$A1$, bcc–$A2$
(metastable)	:	hcp–$A3$

Modelling:

Liquid	:	Substitutional, Redlich–Kister
bcc–$A2$	:	Substitutional, Redlich–Kister
fcc–$A1$	:	Substitutional, Redlich–Kister
hcp–$A3$	:	Substitutional, Redlich–Kister

Assessor and Date:

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Thermodynamic properties of the solution phases (J.mol^{-1})

Phase bcc–$A2$

$$L^{0,bcc-A2}_{Cu,Fe:\square} = 39258.0 - 4.14983\ T$$

Phase fcc–$A1$

$$L^{0,fcc-A1}_{Cu,Fe:\square} = 48232.5 - 8.60954\ T$$

$$L^{1,fcc-A1}_{Cu,Fe:\square} = 8861.88 - 5.28975\ T$$

Phase hcp–$A3$

$$L^{0,hcp-A3}_{Cu,Fe:\square} = 48232.5 - 8.60954\ T$$

$$L^{1,hcp-A3}_{Cu,Fe:\square} = 8861.88 - 5.28975\ T$$

Phase liquid

$$L^{0,liquid}_{Cu,Fe} = 36088.0 - 2.32968\ T$$

$$L^{1,liquid}_{Cu,Fe} = 324.53 - 0.03270\ T$$

$$L^{2,liquid}_{Cu,Fe} = 10355.40 - 3.60297\ T$$

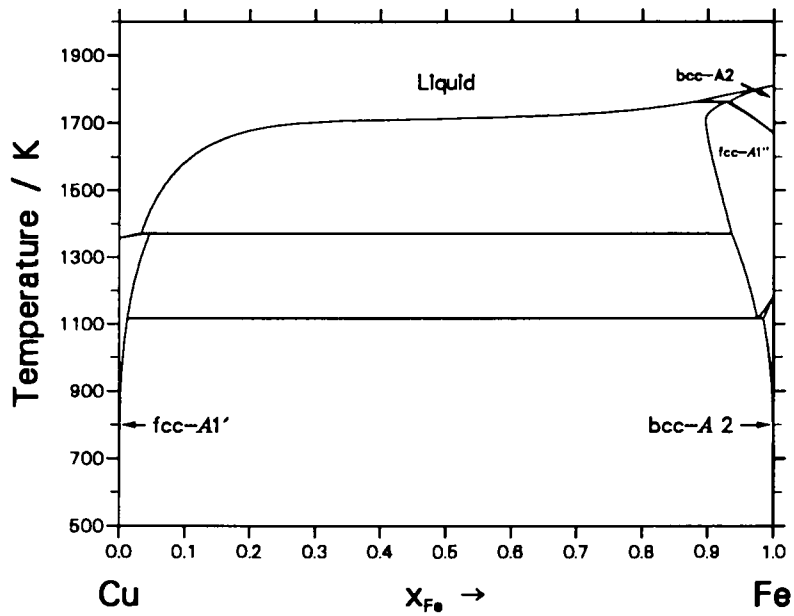

Table I - Invariant Reactions.

Reaction	Type	Compositions x_{Fe}			T / K
Liquid + bcc-$A2$ ⇌ fcc-$A1$"	Peritectic	.880	.933	.927	1762.4
Liquid + fcc-$A1$" ⇌ fcc-$A1$'	Peritectic	.034	.936	.047	1371.4
fcc-$A1$" ⇌ fcc-$A1$' + bcc-$A2$	Eutectoid	.975	.012	.984	1115.9

Table II - Crystal Structure and Phase Description.

System	Phase	Prototype	Pearson Symbol Space Group
Cu-Fe	(Cu)	Cu	*cF4* $Fm\bar{3}m$
	(Fe)	Cu	*cF4* $Fm\bar{3}m$
	(Fe)	W	*cI2* $Im\bar{3}m$